

Uday Singh Kanwar

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Education

McMaster University

Expected Graduation, April 2027

Bachelor of Engineering in Software Engineering

Hamilton, Ontario

- **GPA:** 3.9/4.0- Dean's Honor List, **Faculty of Eng Award of Excellence Scholarship-** \$4000
- **Coursework:** Object Oriented Programming in Java, Software Design, Linux, Git Version Control, CI/CD Pipelines, Digital Logic, Statistics and Probability, Computing in Python, Data Structures and Algorithms.

Work Experience

Software Engineer Intern

Jan 2025 - March 2025

Learning Mode AI- EdTech startup transforming YouTube videos into interactive learning with LLMs.

Remote, ON

- Built **3+** **Go** and **Python** based **RESTful** APIs to enable dynamic quiz generation, concise video summaries, and real-time Q&A by leveraging **Redis** and **GPT-4o** LLM integration.
- Developed **10+** **UI features** and resolved **15+** **bugs**, improving responsiveness by **30%** using **React** and **JavaScript**.
- Worked in a **Docker**-based environment with **Git**, **unit testing**, and docs to deliver reliable, maintainable code.

Machine Learning Engineer

Jan 2025 - Present

McMaster Aerial Robotics and Drone Team: Student Club building drones for SUAS Canada competition.

Hamilton, ON

- Developing real-time **object detection models** using **TensorFlow CNNs** on **NVIDIA Jetson Nano**, achieving over **90%** accuracy in dynamic environments.
- Optimizing AI workflows with **Docker** and **AWS SageMaker**, reducing training and deployment time by 40%.

Full-Stack Developer

Sep 2024 - Dec 2024

Human Endeavour: Organization advancing senior care with smart and assistive technology

Remote, ON

- Managed a **MySQL** database handling real-time data from **15+** **IoT sensors**, improving query performance by **40%** through index optimization and efficient schema design.
- Built an interactive dashboard using **React.js**, a **Flask** backend, and **Tableau** integration, enabling real-time visualization of **500+** daily data points, with front-end workflows prototyped in **Figma** for intuitive UI/UX design.

Embedded Software Engineer Intern

June 2024 - Aug 2024

Human Endeavour

Vaughan, ON

- Developed a **C++** fall detection **algorithm** using an ESP32 micro-controller and accelerometer with **95% accuracy** and implemented an automated email alert service via the **Gmail SMTP server**.
- Implemented a **trilateration algorithm** with Ultra-Wide band technology using **AT Commands** and **C++** for an indoor positioning system with **80% accuracy** in tracking.

Technical Projects

ML-Powered Fraud Detection | *Python, Scikit-learn, Tensorflow, Pandas, NumPy, Matplotlib, Seaborn, Kaggle*

Dec 2024

- Evaluated machine learning models for fraud detection on imbalanced data, applying **feature scaling**, **outlier removal**, and **visualization** to optimize model **performance**.
- Achieved **98.7%** accuracy (oversampling), **94.21%**(undersampling), and **92.1%** using a **TensorFlow Neural Network**.

Bus Forecast at BramHacks | *Typescript, Python, Flask, Google Maps API, Facebook Events API, React.js, Vite*

Nov 2024

- **Won:** Developed a web application with a predictive model for public transportation surges. Used **Google Maps API**, **Facebook Events API**, **Flask** based **Python** backend to **predict** surges.

EcoCoin at Terrahacks | *JavaScript, React, Firebase, CSS, HTML*

Aug 2024

- Developed a **blockchain**-like token system using **Google Cloud**, with **React** for the front end, **Firebase** for backend services, and **Mapbox** for geolocation and map **visualization**.

Hobbies and Clubs | *Google Student Developers' Club, Intramurals: Hockey, Basketball and Volleyball*

Skills

Languages: Python, Java, Go, JavaScript, C++, C, CSS, C, React.js, Typescript

Technologies: AWS, Firebase, Flask, phpMyAdmin, TensorFlow, MySQL, OpenCV, Taipy, Arduino IDE, MatLab, Gitlab, GitHub, AutoDesk Inventor, Maven, Docker